

BioSemi EEG Manual: Boston College Human Neuroscience Facility

Jason Geller, PHD

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SOP

The Human Neuroscience Laboratory houses a 64-channel BioSemi ActiveTwo system from Cortech Solutions.

EEG Room

The Biosemi EEG system is located in McGuinn 405H. Access to these facilities is restricted and must be arranged through Jason Geller, Ph.D.

System

Note

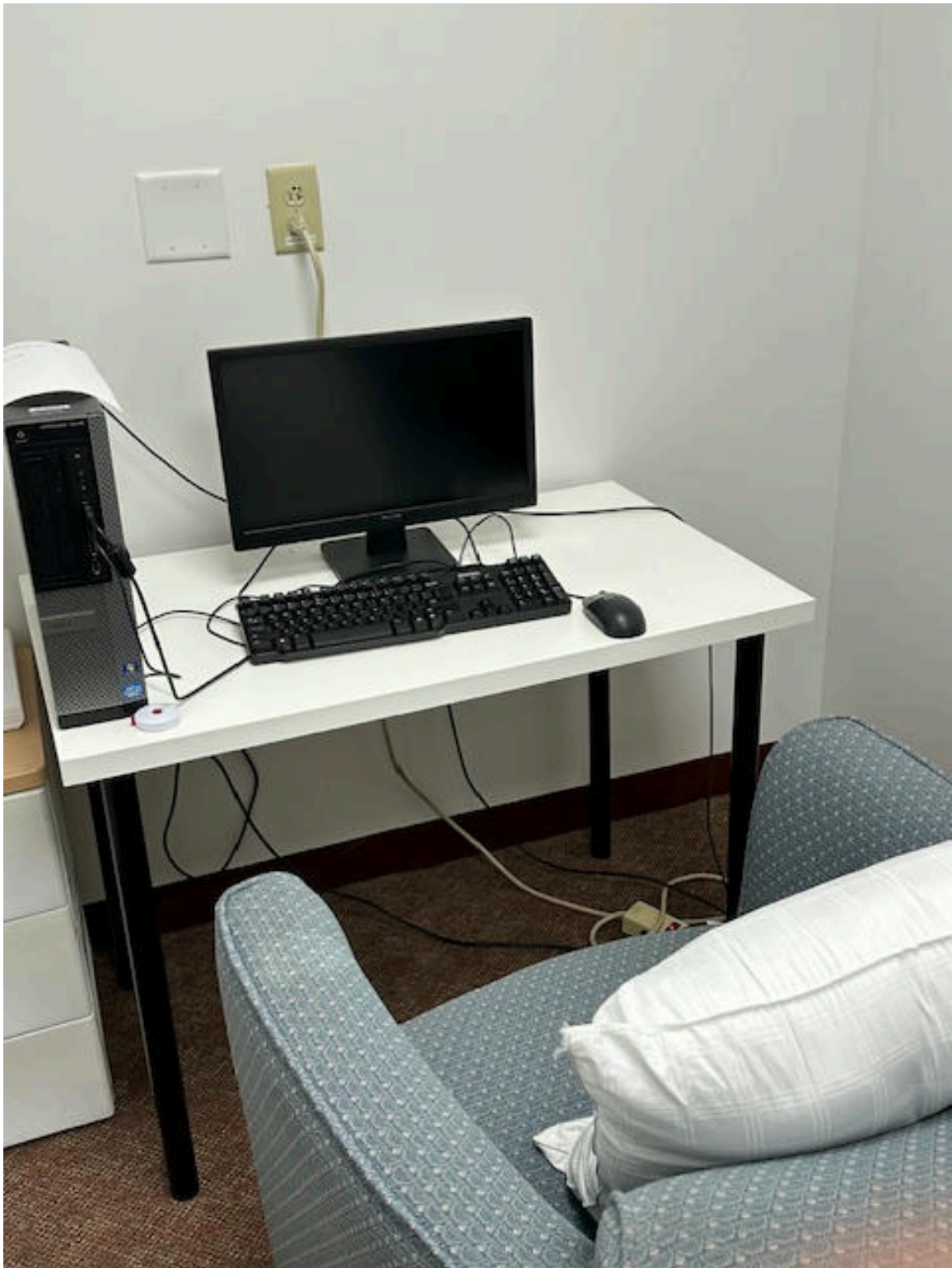
The system can do hyperscanning studies

- This system includes a compact ActiveTwo AD-box, into which sensor-signals attach. In addition to the 64 scalp-signals, the AD-box is also outfitted to receive channels from the face as well as from basic psychophysiological monitors.
- Each AD-box channel consists of a low noise DC coupled post-amplifier, with a first order anti-aliasing filter, followed by a Delta-Sigma modulator with an oversampling rate of 64, and decimation filter with a steep fifth order sinc response and high resolution 24-bit output. The digital outputs of all the AD converters (up to 256) are digitally multiplexed and sent to the PC via a single optical fiber without any compression or other form of data reduction.
- There is a USB2 Receiver which converts the optical data coming from the AD-box to an USB2 output. This receiver also has a trigger port with 16 independent trigger inputs and outputs. This setup keeps the stimulation setup galvanically isolated from the subject. The trigger inputs allow easy setup of ERP measurements and event logging.
- The BioSemi headcaps (we have three different sizes of headcap to accommodate all adults) include the electrode holders. This allows for fast placement of the active electrodes (approximately 15-30 minutes for a 64 channel headcap).
- The BioSemi “Pin-type” Active electrode has been designed for mounting in BioSemi headcap’s. The electrode has a sintered Ag-AgCl electrode tip, providing very low noise, low offset voltages and very stable DC performance. The electrode is completely resistant to long term water and alcohol submergence enabling easy cleaning and disinfecting. The electrodes are numbered with water-proof stickers for easy channel recognition. Because of the Active electrode principle, high electrode impedances do not influence the signal quality.
- One benefit of this system is that it does not require scalp abrasion (and is therefore useful for older adults and children with fragile scalps).

We have adult and child head-caps.

EEG Computers

- **Experiment Computer** (furthest from the door): Used for experiment presentation (e.g., PsychoPy/MATLAB)
 - Equipped with one monitor.
 - Behavioral data is stored on this PC.



- **EEG Computer** (closest to the door): Used for data collection.
 - Equipped with one monitor.

- EEG data is stored on this PC.



- **Login Credentials:**

- **Username:** .\HNL_Lab_405
- **Password:** HN_Lab-mcguinn405

Additional Notes:

- **No Internet Connection:** USB drives must be used to transfer experimental files and store EEG data. An Internet connection can be established if need be.
- **Data Management:** Please transfer EEG data off the system as soon as possible.
- **Material should not be removed from the room.** Bring your own towels/shampoo to use for your testing. We have no washer/dryer available, so you also have to take your towels back home with you for washing/drying
- **Contact Jason Geller when something is missing/broken, or when supplies are running low.**

Who Can Use It?

- Contact the director for the rules and regulations, access to the booking calendar and facilities.
- The BioSemi is available to BC faculty/students and external users
- Usage is not free see (costs)
- To use the system, you need to complete training with director

Support/Training

- **Please note that support generally does not include programming your experiment for you**
 - I have templates for software like PsychoPy that I can provide you to help program your experiment
 - I can provide troubleshooting if experiment is not working properly
 - **Preprocessing/analyzing your data**
 - I have tutorials I can provide you to help with this
 - Consulting is additional cost
 - Helping you design your study, analyze your data, and performing analyses/write-up results is contributing to your project under the credit taxonomy <https://credit.niso.org/> and will warrant authorship (REGARDLESS IF YOU ARE PAYING ME)
 - Please keep this in mind when reaching out for help.
- **Cleaning the equipment after your done**
 - Cleaning up after the experiment is your responsibility! My job is not to clean up after users.

Booking

- Book the lab via iLab (email John Maloney (<malonegl@bc.edu>), if you don't have access.) Make sure that the lab booking is booked under

64-channel BioSemi ActiveTwo system from Cortech Solutions, 405H

once booked, the time should appear on the calendar. If not, you may have booked it under another calendar by mistake, in which case edit it to make sure other people will be able to see that the room is booked.

- Before starting a new study, contact the director and provide them with details of your study:
 - When do you plan on running experiments?
 - How many participants will you test?
 - Who is running the study (including PI, experiment leader, master thesis students, and/or interns)? Also provide their contact information.
 - Do you have approval from an ethical committee to conduct this EEG study (+ add ethical approval code)?
 - Do you have prior experience with the BioSemi?

Costs

- You pay each hour you test
 - Self Use Sun-Sat 12:00 AM - 12:00 AM (Internal: \$14.78/hr, External; Non-Academic: \$14.78/hr, External Academic: \$14.78/hr) (Trained)
 - Assisted Use Mon-Fri 09:00 AM - 05:00 PM (Internal: \$102.92/hr; External Non-Academic: \$102.92/hr, External Academic: \$102.92/hr) (All Customers) (needs approval)
 - Training Session Mon-Fri 09:00 AM - 05:00 PM (Internal: \$102.92/hr; External Non-Academic: \$102.92/hr, External Academic: \$102.92/hr) (All Customers) (needs approval)

File Storage

- On the desktops of both PCs, you will find a folder “Experiments”. Create a subfolder with your name and store all your files here. • Always copy your data to an external device and remove it from the PC.

! Important

- The PC's hard disks have limited space and will be cleaned regularly

Cleaning

- You can only use the BioSemi after following the BioSemi EEG workshop, during which you will learn how to thoroughly clean the BioSemi subparts.

- Researchers exempted from this training still need to obtain instructions from the director on how to clean the equipment, so the same procedure is always used for cleaning by all users.
- The electrodes are extremely fragile. It is therefore crucial to treat them with great care and clean them exactly as instructed.
- Your access to the equipment may be revoked if the cleaning is not satisfactory.
- Please bring your own towels and take them back home with you for washing/drying. Do not leave dirty towels in the lab!
- If you want to offer water to the participant bring your own cups and clean them afterwards. Do not leave dirty cups in the lab!

DOs and DON'Ts

Dos

- **Always keep one battery charging.** The batteries break if you fully discharge them (“deep discharge”). You cannot harm the batteries by charging them too much.
- Inform the director immediately if something seems to be broken or needs restocking (gel, syringes, adhesive disks , etc.). The EEG equipment is expensive, so handle it with great care!
- Always book all your test slots via iLab. This includes booking time to pilot your experiment.

Donts

- Don't change any hardware or permanent software settings (e.g., refresh rate of the monitor) in our set-ups without asking the director.
- Don't get the connectors wet! Make sure during cleaning that **only the caps and the electrodes get wet**. Make sure that the connectors don't get wet by wrapping them carefully in a towel or plastic bag.
 - If you need a cap that is currently wet you can use a blow dryer, but use cold air only!
- Don't leave your data on the computer. Back-up and remove your data from the acquisition machines after each session (e.g., at the end of the day). Computers will regularly be cleaned and loss of data is your own responsibility.

User Guide

Before the Participant Arrives

1. Schedule in iLab!

- Users must reserve the EEG to use it.

Human Neuroscience Laboratory

[About Our Core](#) [Schedule Equipment](#) [Staff Only](#) [Staff Only](#) [R](#)

[Schedule Resources](#) [Go to Kiosk](#) [Timeline View](#)

To make a reservation:

1. Click on the category name to expand the categories
2. Click on the instrument name or on the 'View Schedule' button on the far right.
3. On the calendar interface click and drag to reserve a time slot for the day and time desired
4. In the 'use and cost of reservation' section, use the drop down under *Pricing Details* button to select reservation type.
5. Click on the 'Save' button

Please be advised the calendars are for approved users only.
If you are requesting staff assistance or instrument training, after you create your reservation you will be notified via email if your reservation is approved/disapproved marked as 'non spam' to ensure they go to your inbox. Thanks!

64-channel BioSemi ActiveTwo system from Cortech Solutions, 405H [description](#) [pricing](#)

The laboratory houses a 64-channel BioSemi ActiveTwo system from Cortech Solutions. This system includes a compact ActiveTwo AD-box, into which sensor-signals attach. In addition to the 64 scalp-signals, the AD-box is channels from the face as well as from basic psychophysiological monitors. Each AD-box channel consists of a low noise DC coupled post-amplifier, with a first order anti-aliasing filter, followed by a Delta-Sigma modulator with a decimation filter with a steep fifth order sinc response and high resolution 24-bit output. The digital outputs of all the AD converters (up to 256) are digitally multiplexed and sent to the PC via a single optical fiber without any data reduction.

There is a USB2 Receiver which converts the optical data coming from the AD-box to an USB2 output. This receiver also has a trigger port with 16 independent trigger inputs and outputs. This setup keeps the stimulation set up the subject. The trigger inputs allow easy setup of ERP measurements and event logging. The BioSemi headcaps (we have three different sizes of headcap to accommodate all adults) include the electrode holders. This active electrodes (approximately 15 minutes for a 64 channel headcap).

IF YOU ARE SCHEDULING USAGE FOR HUMAN SUBJECT RESEARCH PURPOSES, PLEASE ENSURE ALL PROCEDURES ARE IRB-APPROVED

2. Login to iLab

- There is a new kiosk system in place where in order to turn on the equipment you need to start your session.
 - You can do so by navigating to Go to Kiosk under scheduling.
- Find your reservation and start your session.

[Start Walkup Session](#)

[Refresh Sessions](#)

All Kiosk Sessions ☐ Show completed sessions

Time Scale: -6 hours to 6 hours

[Clear](#)

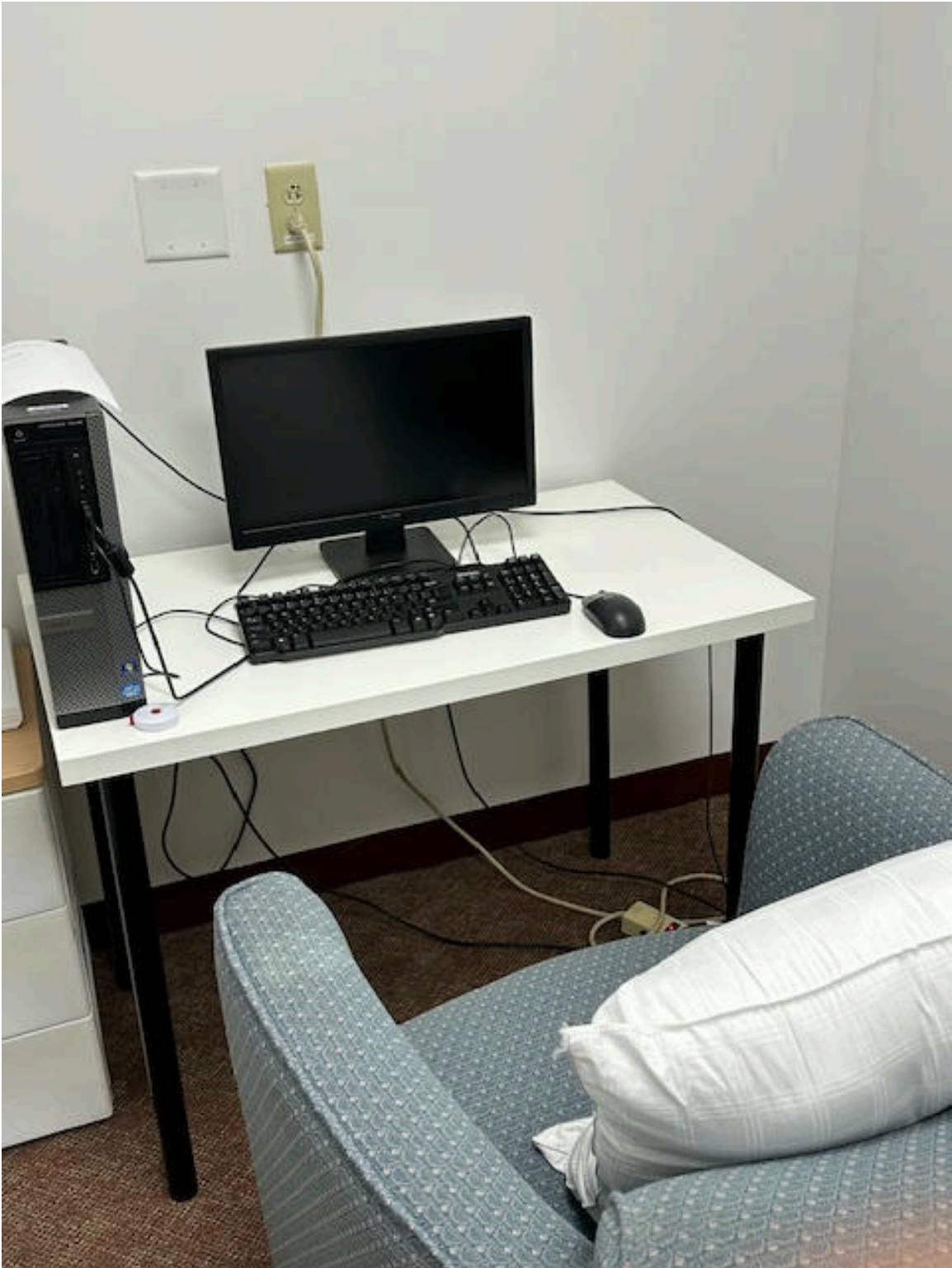
Owner	Resource	Start	End	Actions
Carolina Alvarez	Tobii Pro Spectrum system 1, 405G (Tobii Pro Spectrum system 1)	03 Oct 09:50 AM	2 hours left	Finish
Stephanie M Nemeh	Tobii Pro Spectrum system 1, 405G (Tobii Pro Spectrum system 1)	03 Oct 12:00 PM	03 Oct 01:30 PM	Details
Carolina Alvarez	Tobii Pro Spectrum system 1, 405G (Tobii Pro Spectrum system 1)	03 Oct 01:30 PM	03 Oct 04:30 PM	Details
Sandry Garcia	The Magstim Rapid ² repetitive transcranial magnetic stimulator, 405A (The Magstim Rapid ² repetitive transcranial magnetic stimulator)	03 Oct 03:00 PM	03 Oct 05:30 PM	Details

3. Switch on both PCs

- Computer with Actiview



- Display PC:



4. Switch on the AD box (On/Off button).

Battery check



Figure 1. AD box on top of battery



Figure 2. Battery charging

5. Check the battery

Replace it if necessary and ensure one battery is always charging.

6. Prepare all necessary materials on the EEG table, away from the computers.

- This is generally done outside the room.

7. Fill two syringes with gel (clean the tip of the tube before and after).
dry, but use cold air only, do not to damage the cap.



Figure 3. Fill the syringes with gel



Figure 4. Clean the tip of the gel tube

8. Take out gauze pads and alcohol spray bottle
9. Order and untie the electrodes.
10. Ensure the head cap is dry (use the cap dryer if needed).

Upon Participant Arrival

1. Welcome & Informed Consent

- Greet the participant warmly and reassure them that the procedure is safe and non-invasive.
- Explain that this may be their first EEG experience, so you will go slowly and describe each step in detail
- Provide the **information sheet** and give the participant time to read it carefully
- Answer any questions they may have.
- Ask the participant to sign the **informed consent form** before proceeding

2. Remove accessories

- Ask the participant to remove **earrings, hairpins, hair elastics, headbands, and glasses** if possible.
- Explain that this ensures proper electrode placement and avoids interference.

3. Preparing the scalp

- Put on gloves.
- Spray **70% isopropyl alcohol** onto a piece of sterile gauze.
- Gently wipe the participant's scalp where electrodes will be placed (especially the frontal hairline and crown)
 - Explain that this helps remove oils and hair products, improving signal quality.

4. Measuring Head Circumference

- Take a **measuring tape** (centimetre side).
- Place the tape at the **nasion** (the small indentation between the eyes at the top of the nose).
- Wrap the tape around the participant's head, passing just above the ears and around the back of the head.
- Record the measurement to determine the correct **EEG cap size** (tag on each cap indicates size)

5. Selecting and Placing the Cap

- Choose the EEG cap size according to the **tag label** (e.g., 54 cm, 56 cm, etc.).
- Explain to the participant that the cap will feel snug but should not be painful.
- Position the cap on the head, aligning the **centre front electrode** (usually Fpz) with the nasion.
- Ensure the cap is symmetrical and evenly placed.
 - Cz should be at Vertex (top of head)
 - Measure From Nasion to Inion
 - Measure between left and right Preaurical points
 - Fp1/2 should be ~10% higher than Nasion (Distance from nasion to inion is usually around 35-40 cm, so ~ 4 cm)
 - O1/2 should be ~ 10% higher than Inion

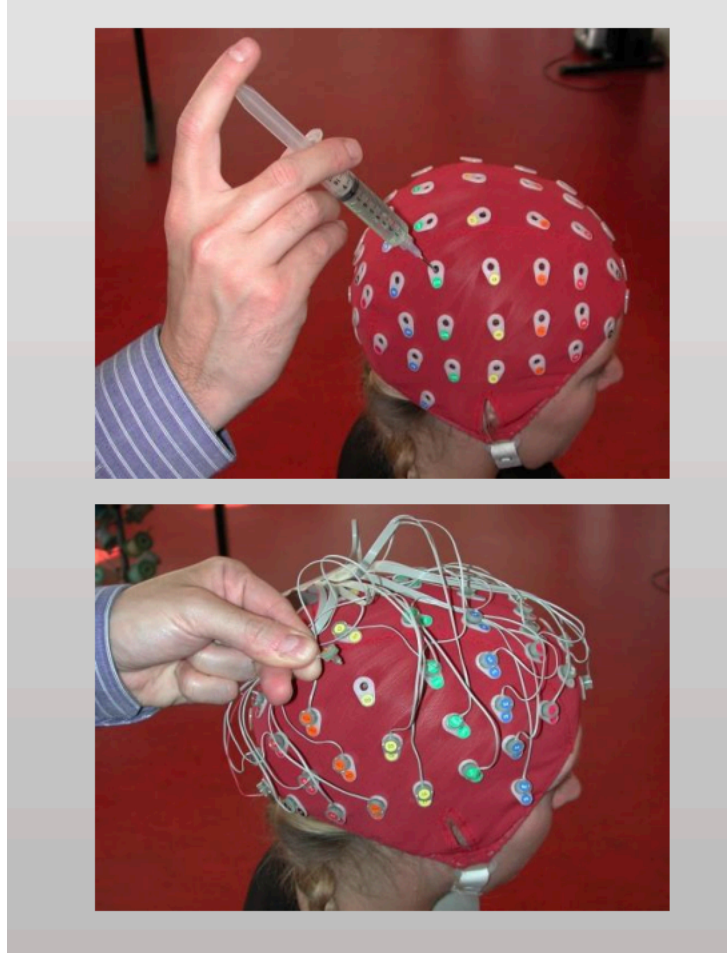
T7/8 should be ~10% higher than Preaurical points

{fig-align="center"}

5. Gel

- Squeeze in a small amount of gel while moving the syringe upwards.
- Gently fiddle around with the syringe first, as this moves hair aside. Ask the participant for feedback to avoid scratching their scalp.
- Too much gel can create “bridges” between individual electrodes. Be especially careful around the CMS/DRL electrodes as these are crucial for proper data collection.
 - It is easier to add extra gel later on than to remove excess gel.

- You may notice that you easily lose track of which electrode gaps you have already filled. It can help to come up with a system (e.g., start from the back, or if you are with two experimenters you can each start from a side and work your way to the middle).



7. Electrode Placement

- Electrodes damage easily, so always handle them with great care.
- Electrodes are numbered. Place electrodes in ascending order to avoid tangling the wires. Start at the base of the skull and work your way towards the forehead. Make sure the wires end up in the participant's neck.
- Tape the electrodes to the participant's shoulder to avoid putting stress on the cables.
- Wrap the CMS/DRL around the electrode wires 4-5 times to limit noise.
- Connect the electrode bundles to the AD box.

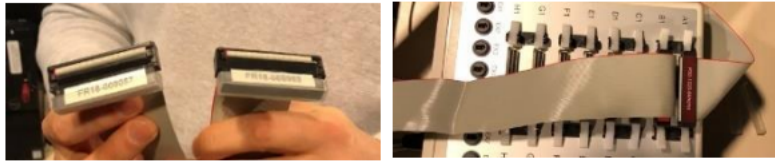


Figure 8. Connect the electrode bundles to the AD box

i Note

An unfortunate problem in the EEG world is that not all hair types work with EEG. If your participant has thick/dense/curly hair or special hairstyles like braids you probably won't be able to perform EEG.

Research project from the BRAIN Lab at the University of Central Florida about the effect of hair on EEG data collection.

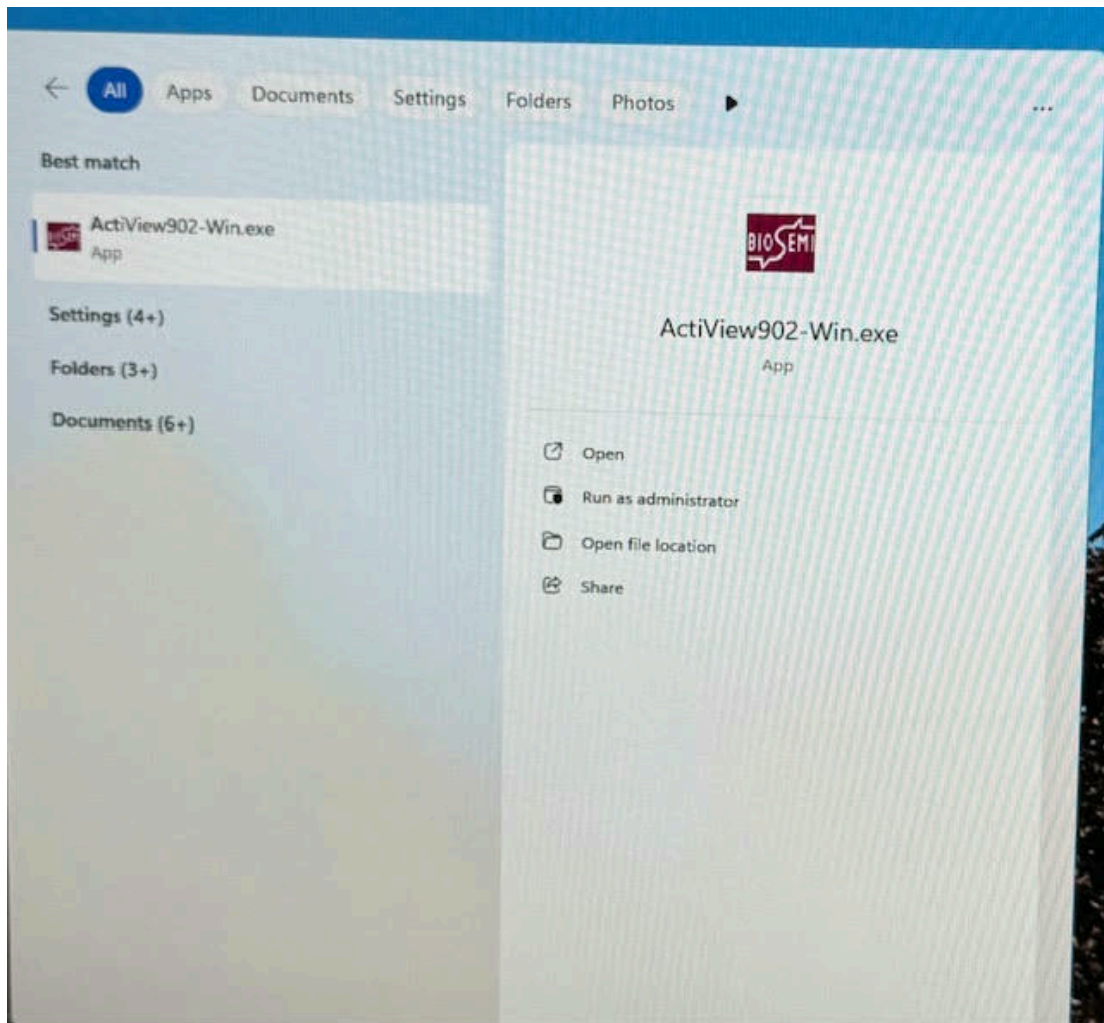
Choy, T., Baker, E., & Stavropoulos, K. (2022). Systemic Racism in EEG Research: Considerations and Potential Solutions. *Affective Science*, 3(1), 14–20. <https://doi.org/10.1007/s42761-021-00050-0>

Manfredi, M., Comfort, W. E., Marques, L. M., Rego, G. G., Egito, J. H., Romero, R. L., & Boggio, P. S. (2023). Understanding racial bias through electroencephalography. *BMC Psychology*, 11(1), 81. <https://doi.org/10.1186/s40359-023-01125-2>

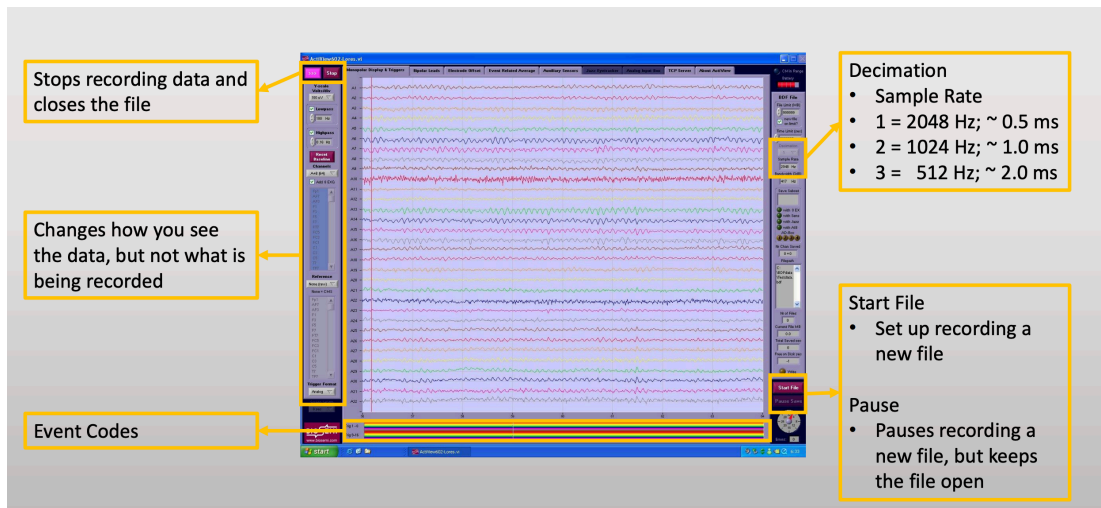
Penner, F., Wall, K. M., Guan, K. W., Huang, H. J., Richardson, L., Dunbar, A. S., Groh, A. M., & Rutherford, H. J. V. (2023). Racial disparities in EEG research and their implications for our understanding of the maternal brain. *Cognitive, Affective, & Behavioral Neuroscience*, 23(1), 1–16. <https://doi.org/10.3758/s13415-022-01040-w>

Start ActiView

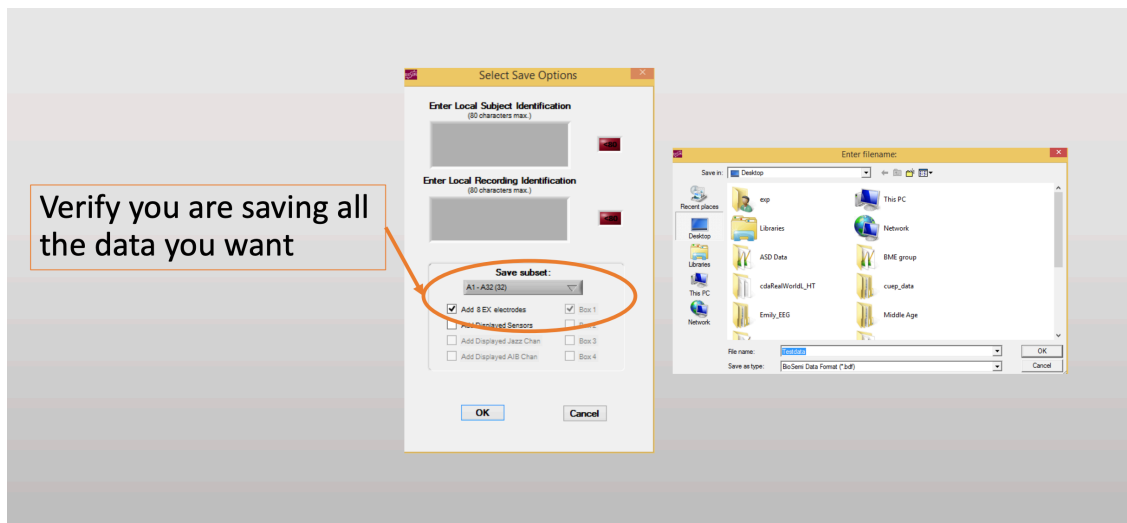
- Start ActiView
 - On the data acquisition computer, navigate to Actiview902-win.exe in the search bar.



- ▶ When you open it up you should see a screen that looks like the one below (please note in new version of ActiView event codes need to be changed in the About Actiview tab).



- Start the recording
- Set up the configuration file for 64 channels
 - This is done by default.cfg that I have loaded on the computer (you should not have to do anything)
- Start the file and enter a file path for saving the data.



- Check the battery status in ActiView.
- Adjust settings such as decimation and trigger format (About tab)
- Check the impedance of the electrodes and make necessary adjustments.
 - Switch to tab Electrode Offset.
- Change the displayed range scale to 50 μ V.

- The impedance of each electrode must be between $-20\ \mu\text{V}$ and $20\ \mu\text{V}$. Adjust electrodes that are not within these values by adding/ removing some gel, moving hair below the electrode or wiggling the electrode.

example code on page 15.



Figure 9. Set-up your file in ActiView

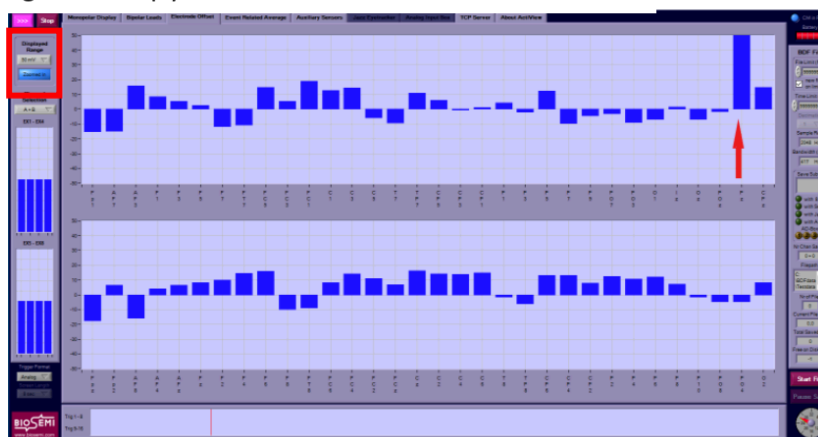


Figure 10. The *electrode offset* tab.

- If the electrode is still far outside the $[-20\ 20\ \mu\text{V}]$ range, the electrode might be broken and needs to be replaced.
- In case this electrode is not crucial for your study you can write down the code of the electrode (e.g., Fz)
- A bad connection of the reference electrode results in a bad signal for all electrodes. Check whether the DRL and CMS electrodes are placed correctly and provide a good signal.

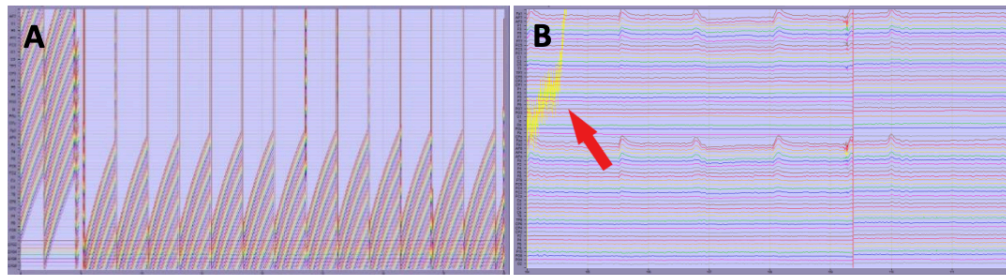


Figure 11. (A) This pattern typically results from a bad connection of the DRL or CMS electrodes. (B) This pattern typically results from the cap label not sticking out of the cap.

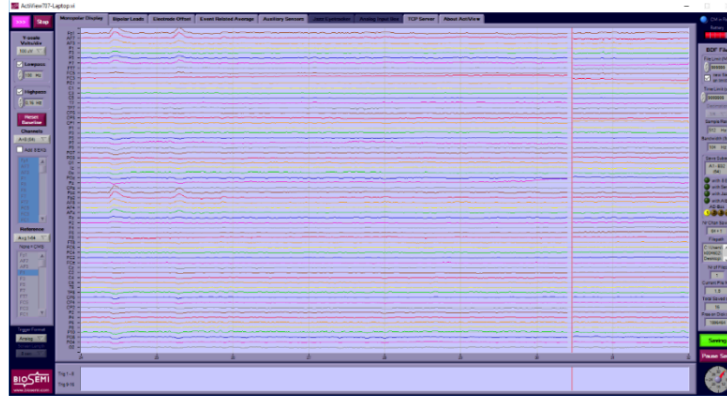
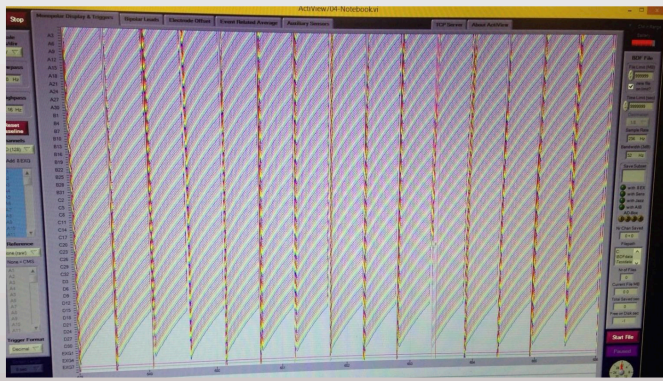


Figure 12. Ideally your data should look something like this, 100Hz sampling rate.



Almost always the solution is to add more gel to CMS / DRL

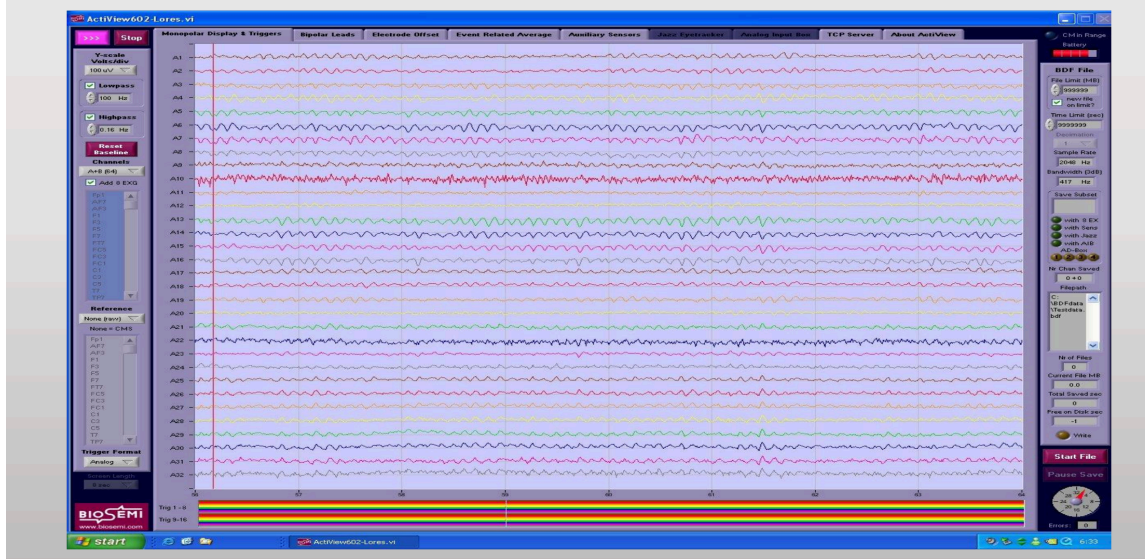
**** Could also indicate a bad electrode. ****



Will be blinking

- The cap-label not sticking out of the cap results in a typical signal distortion
- Show the participant what happens when they blink, cough, swallow, grind their teeth, etc. Instruct the participant to minimize these movements during the experiment.
- If all electrodes are connected correctly, ActiView should look like this:

What you want to see



- Before recording, change the Trigger Format to decimal so you can see the triggers you are recording in real time.

Start A Recording

- Start recording (check that the saving button turns green).

- This is at the top in the new version of ActiView.

• File is setup, but not saving

• File is saving

Manually start or stop saving the EEG

- 2. Dim the lights before starting the experiment.**
- 3. Adjust lights during breaks to increase alertness.**
- 4. Start the experiment.**

After the Experiment

- 1. Make sure you stop the recording when finished (the top left button in ActiView)**
- 2. Disconnect the cap from the amplifiers and move the participant back into the prep area.**
- 3. Gently remove electrodes from cap**
- 4. Provide the participant with the option to wash his/her hair and face in the sink provided in the lab.**
 - They may also use the restroom that is located on the 4th floor. You may offer them a towel, shampoo, and conditioner.
- 5. After the subject has cleaned up, debrief him/her and thank the participant for coming.**

After the participant leaves

- 1. Cleaning electrodes**
 - Do not get the connectors wet
 - Do not let the electrodes touch metal (reduces life span)
 - Rinse in warm water / under the facet
 - Wipe of remaining gel (fingers, cotton swab, Q-tip)
 - Pat dry with a towel and store in a cool dark place



2. Cleaning the cap

- The cap should soak in the warm soapy water for approximately 10 minutes.
- Remove cap from soapy water and rinse. Use the water-pik to clean all the cap electrodes.
- Spray the cap with Envirocide and let sit for 3 minutes, then rinse. Meanwhile, soak facial electrodes in ethyl alcohol for 3 minutes.

When cap is clean, stuff it with a small towel to make a head shape and let it dry on the counter.

- Optionally we have a cap dryer you can use.

3. Clean up and store all equipment.

Experiment Stuff

- Check out some Python template experiments to help you get started (<https://osf.io/m7ec4>)
- A student of mine and I created a Google Colab for project highlighting preprocessing pipeline I use: https://colab.research.google.com/drive/17Td09p3aZcHQDa2AgZ_fSI38cLKFfn8j#scrollTo=b6Lq89RuDhma